1 Adam Results

Table 1: Mean ± std accuracy across 3 runs per configuration.

Dataset	# Train	Size	FFT	LoRA Rank r								
				1	2	4	8	16	32	64	128	256
ARC Easy	2,251	0.5B	61.3±1.7	64.5±0.2	65.3±0.3	65.8±0.6	65.3±0.3	66.6±0.7	66.1±0.9	64.7±1.0	65.7±1.1	65.4±1.1
		1.5B	74.8±1.1	75.9±0.8	76.8±0.6	77.4±1.4	77.7±1.2	78.3±0.7	78.3±0.5	78.8±0.3	78.4±1.4	79.5±1.0
		3B	78.8±1.9	78.0±0.3	78.6±0.7	79.4±1.4	79.8±0.3	80.2±0.1	79.9±0.9	80.7±0.7	80.2±0.8	81.1±1.0
ARC Chall.	1,119	0.5B	30.1±1.2	31.9±2.1	31.4±1.3	31.0±0.9	31.4±0.6	31.8±1.4	31.5±0.5	31.7±0.7	29.7±0.8	32.1±1.4
		1.5B	43.8±0.9	40.9±0.2	40.8±0.5	41.3±0.5	42.3±1.0	42.8±1.1	42.9±1.8	42.8±0.6	42.5±1.3	43.2±0.5
		3B	46.5±0.9	44.7±0.3	44.6±0.3	44.1±0.4	45.8±0.8	46.7±1.3	47.4±0.8	48.9±1.0	47.2±2.6	47.5±1.3
CSQA	9,741	0.5B	60.5±2.5	66.3±0.4	65.9±0.9	64.3±1.1	66.0±0.7	66.3±0.4	66.1±0.3	65.2±1.1	65.6±1.1	64.6±0.7
		1.5B	72.7±0.7	80.5±1.1	79.8±0.2	80.3±0.5	80.1±1.1	80.4±0.6	79.7±0.4	79.6±0.9	80.1±0.6	78.9±0.8
		3B	77.7±1.6	82.7±0.9	82.6±0.7	81.6±0.5	82.1±0.8	83.0±0.9	82.5±0.2	81.9±0.3	82.4±0.7	82.1±0.5

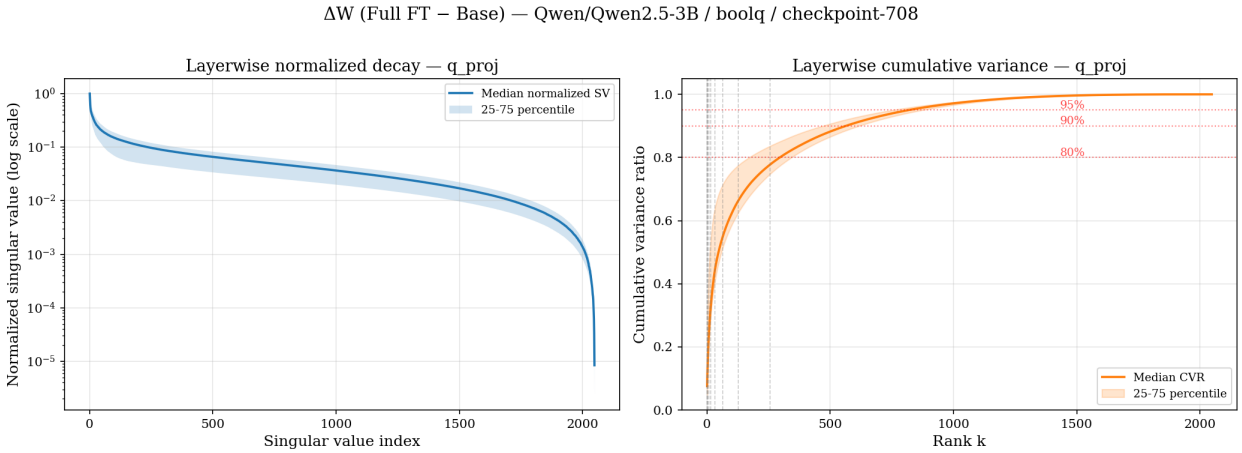

Figure 1: SVD of ΔW (Full FT − Base) — BoolQ (lr=2e-5), Q projection.

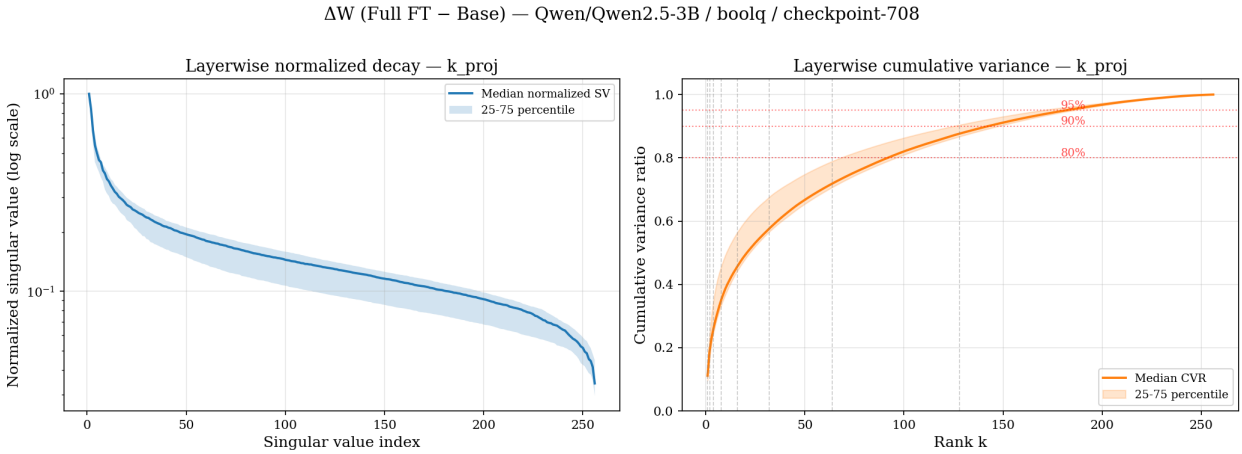

Figure 2: SVD of ΔW (Full FT − Base) — BoolQ (lr=2e-5), K projection.

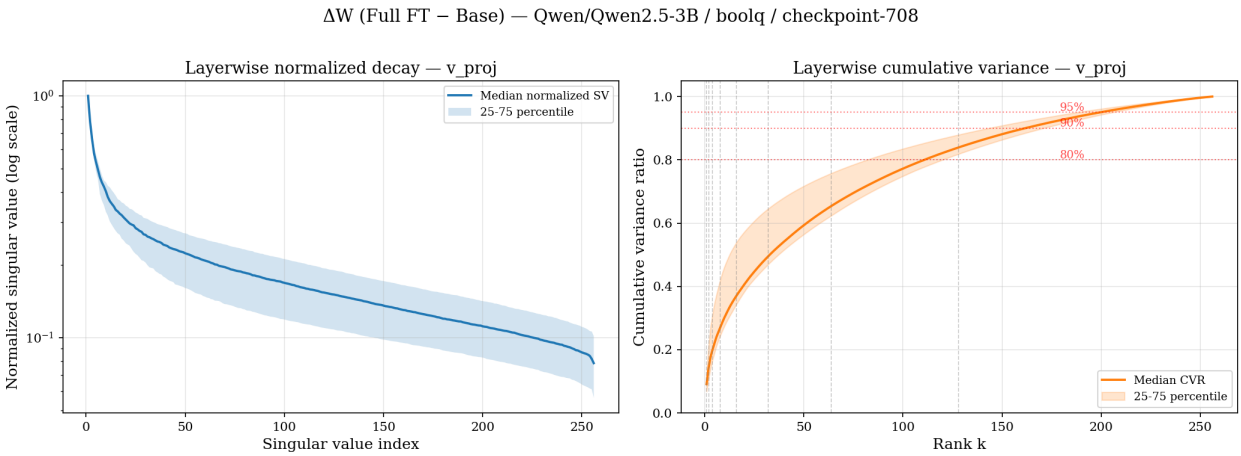

Figure 3: SVD of ΔW (Full FT − Base) — BoolQ (lr=2e-5), V projection.

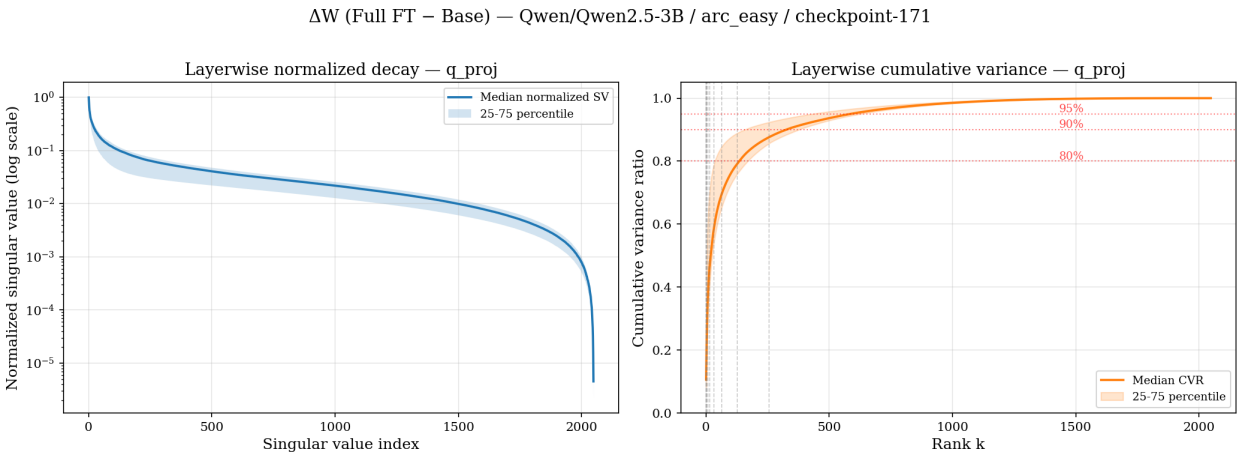

Figure 4: SVD of ΔW (Full FT − Base) — ARC-Easy (lr=2e-5), Q projection.

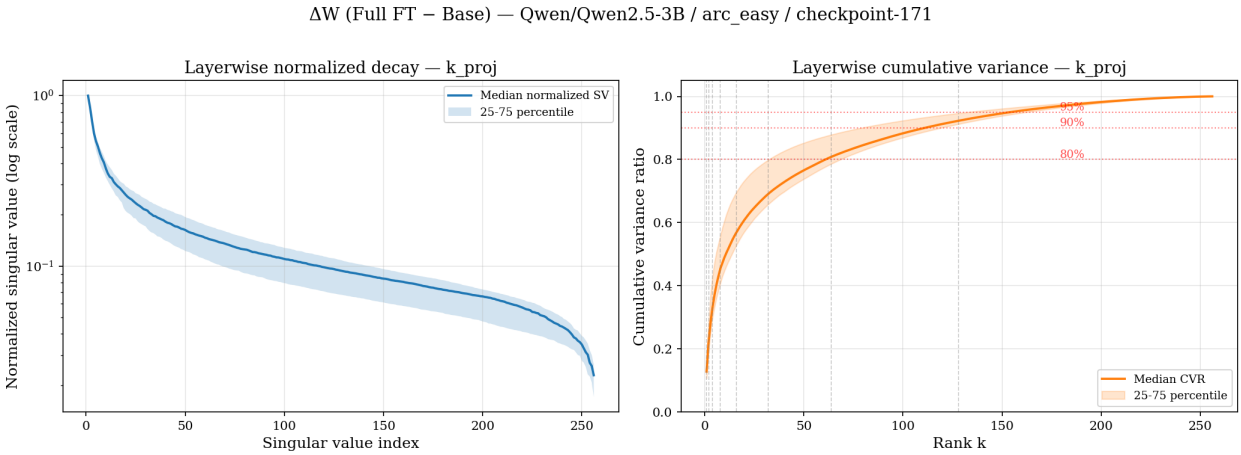

Figure 5: SVD of ΔW (Full FT − Base) — ARC-Easy (lr=2e-5), K projection.

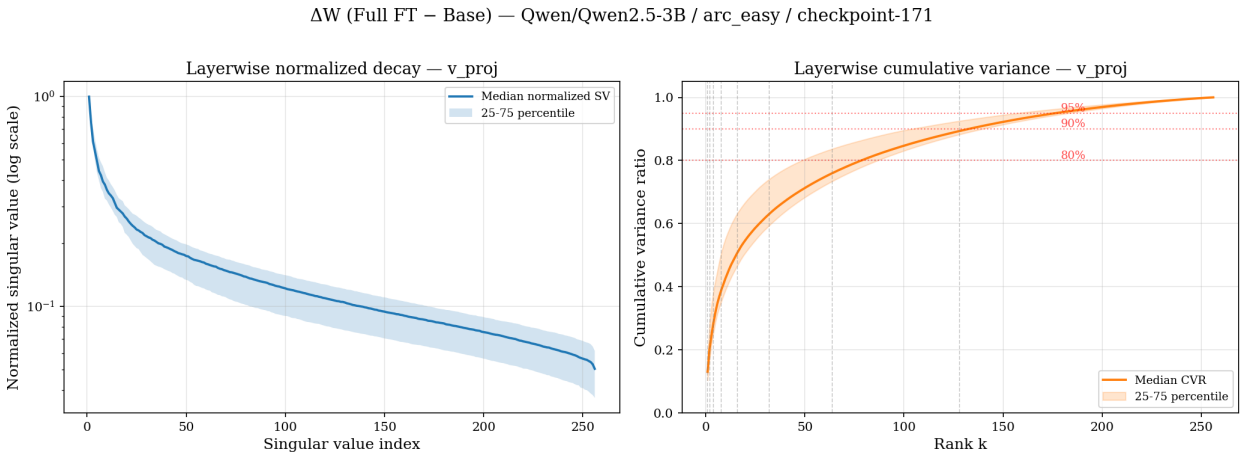

Figure 6: SVD of ΔW (Full FT − Base) — ARC-Easy (lr=2e-5), V projection.

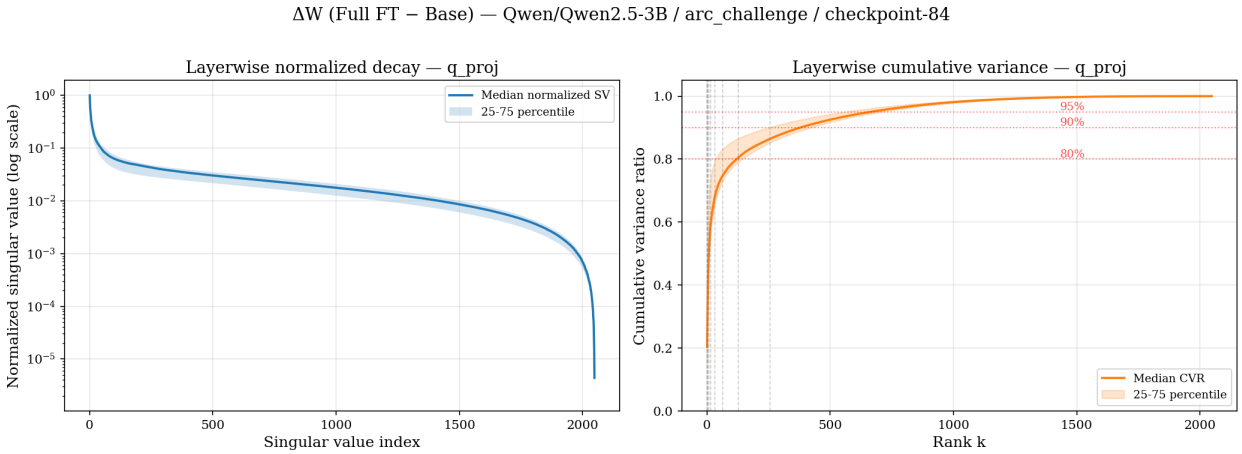

Figure 7: SVD of ΔW (Full FT − Base) — ARC-Easy (lr=2e-5), Q projection.

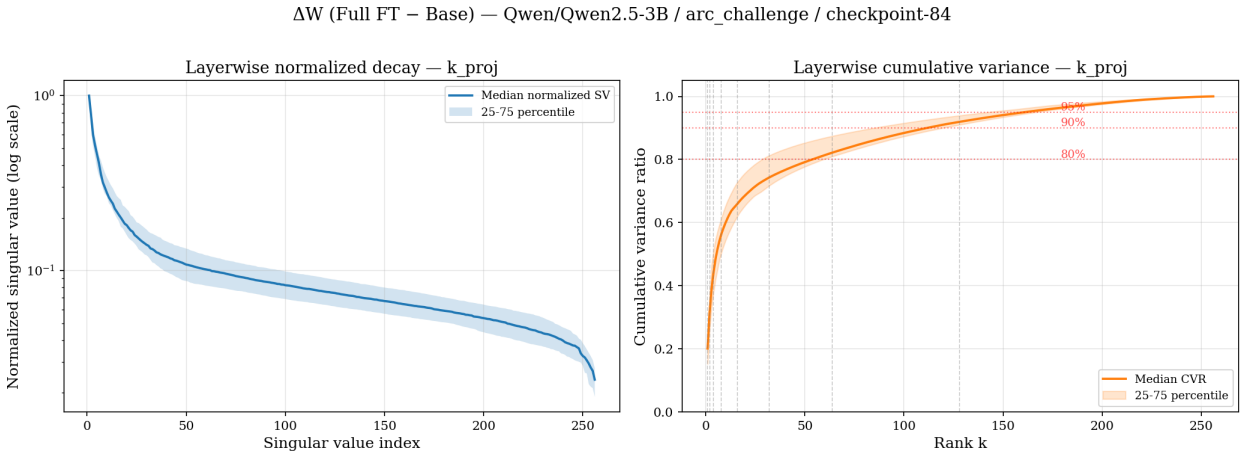

Figure 8: SVD of ΔW (Full FT − Base) — ARC-Easy (lr=2e-5), K projection.

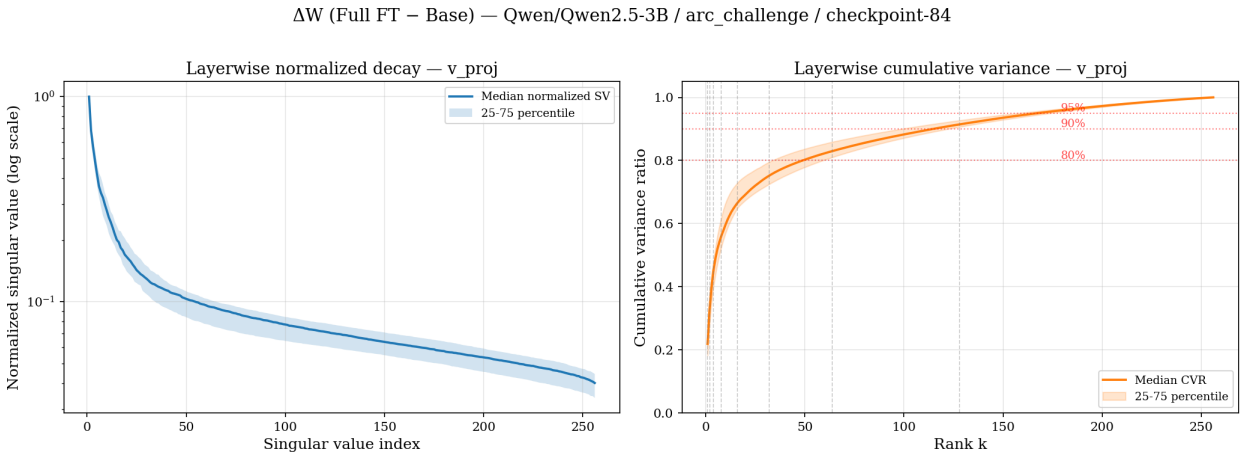

Figure 9: SVD of ΔW (Full FT − Base) — ARC-Easy (lr=2e-5), V projection.

2 SGD with Momentum Results

Table 2: Full fine-tuning results on Qwen2.5-3B with SGD.

Dataset	LR	Epochs	Batch	Accum	Momentum	Scheduler	Warmup	Acc (norm)
ARC-Easy	1e-4	4	4	4	0.9	cosine	0.2	80.5
ARC-Challenge	1e-4	4	4	4	0.9	cosine	0.2	50.0
BoolQ	1e-3	4	4	4	0.9	cosine	0.2	86.5

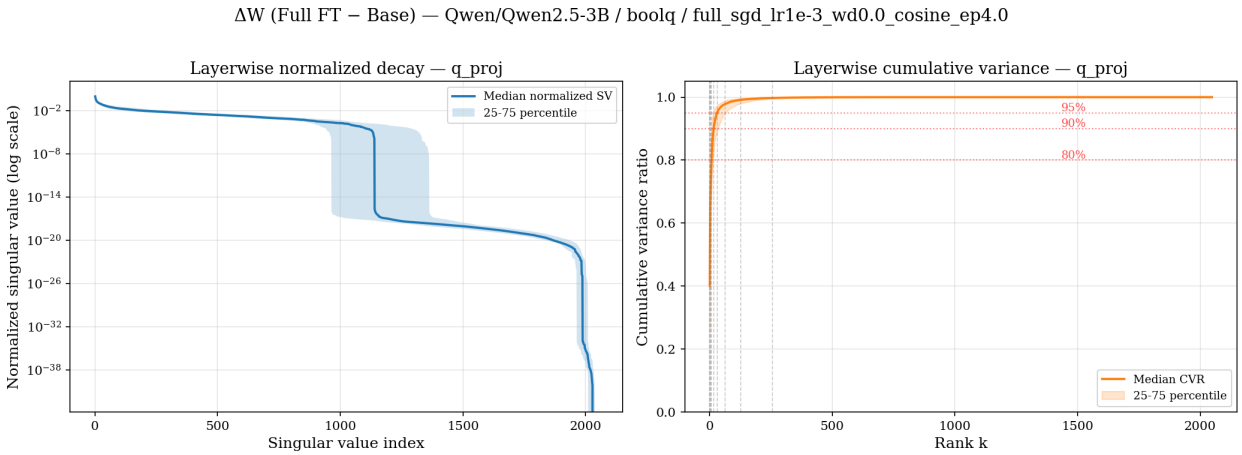

Figure 10: SVD of ΔW (Full FT − Base) — BoolQ (lr=1e-3), Q projection.

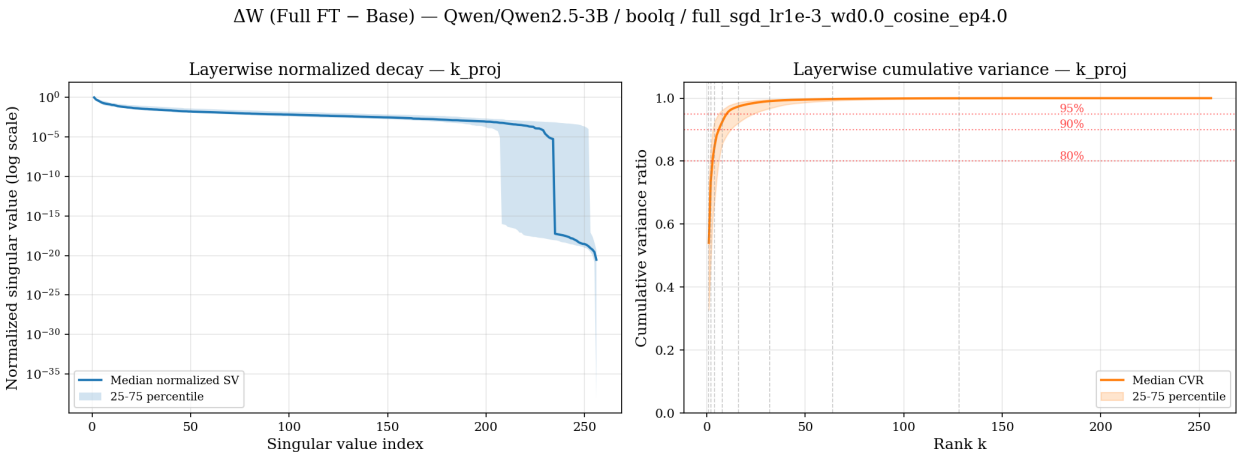

Figure 11: SVD of ΔW (Full FT − Base) — BoolQ (lr=1e-3), K projection.

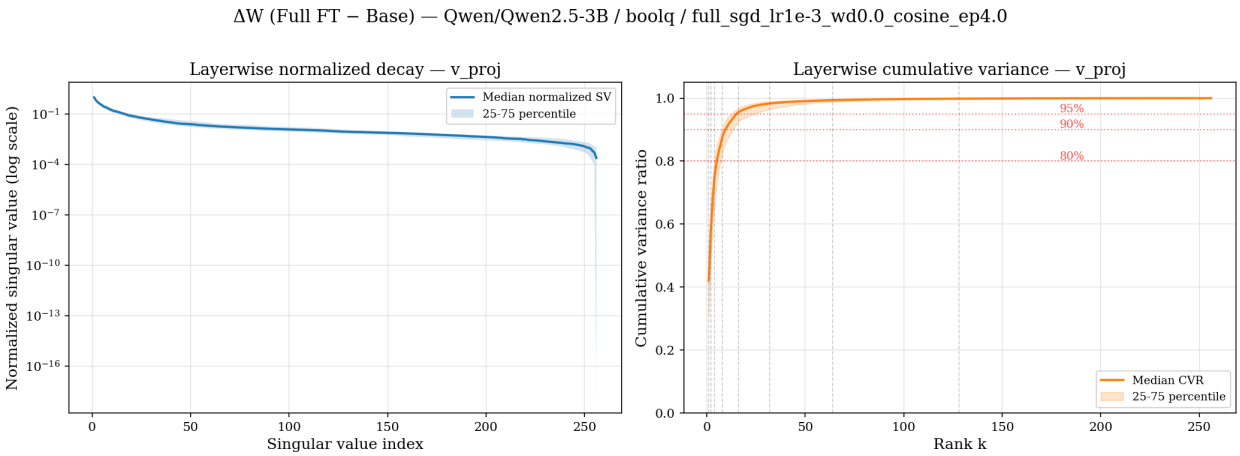

Figure 12: SVD of ΔW (Full FT − Base) — BoolQ (lr=1e-3), V projection.

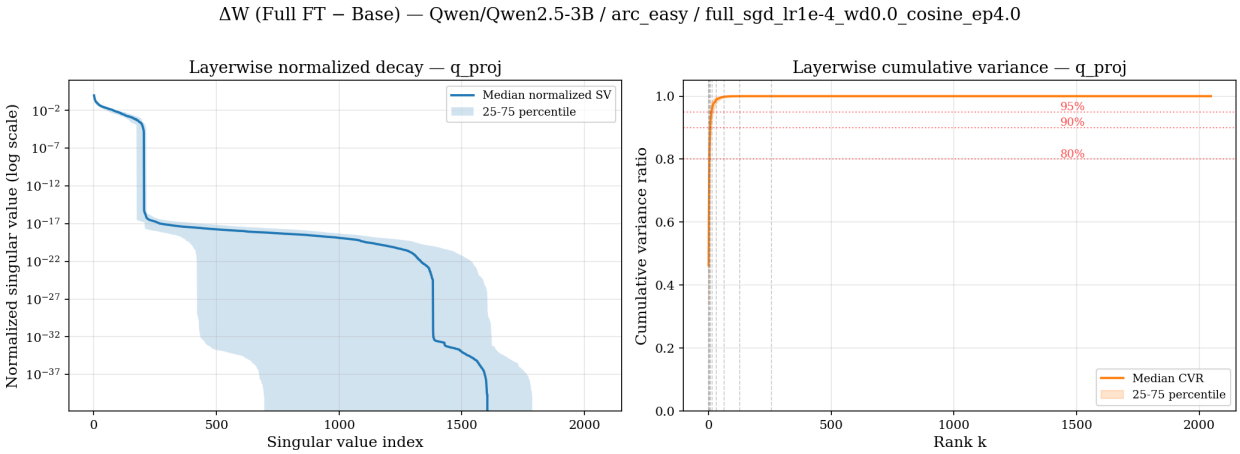

Figure 13: SVD of ΔW (Full FT − Base) — ARC-Easy (lr=1e-4), Q projection.

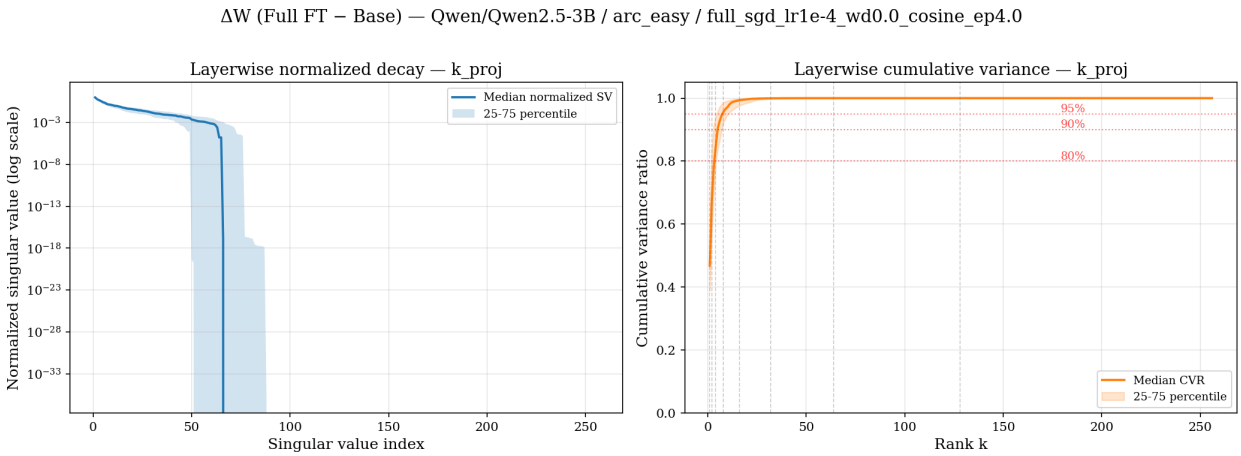

Figure 14: SVD of ΔW (Full FT − Base) — ARC-Easy (lr=1e-4), K projection.

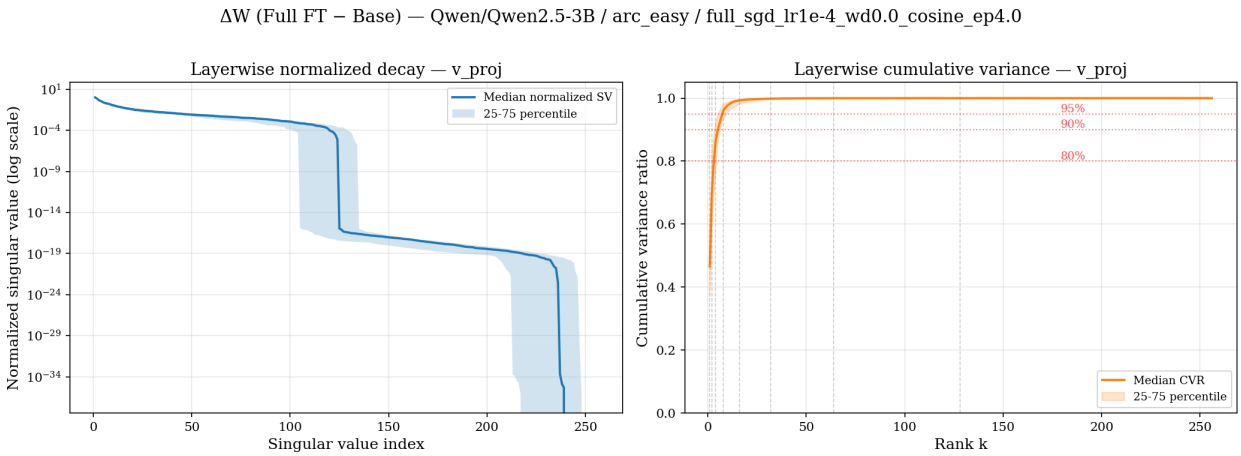

Figure 15: SVD of ΔW (Full FT − Base) — ARC-Easy (lr=1e-4), V projection.

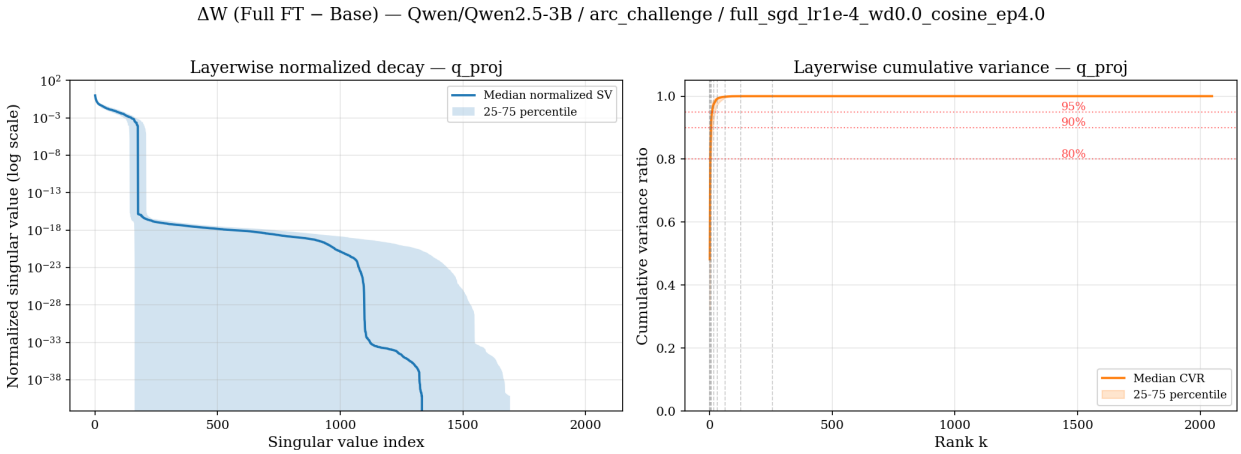

Figure 16: SVD of ΔW (Full FT − Base) — ARC-Challenge (lr=1e-4), Q projection.

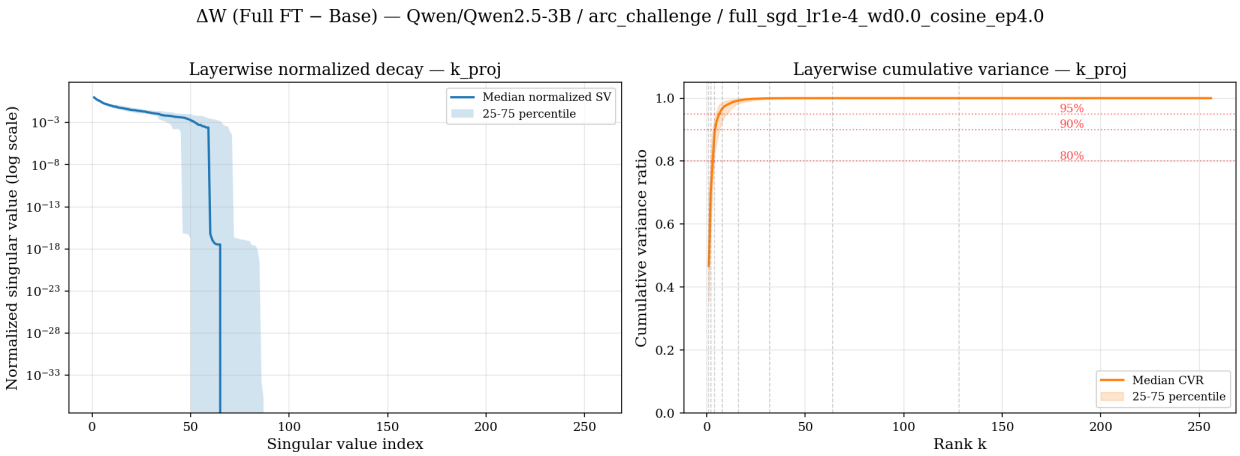

Figure 17: SVD of ΔW (Full FT − Base) — ARC-Challenge (lr=1e-4), K projection.

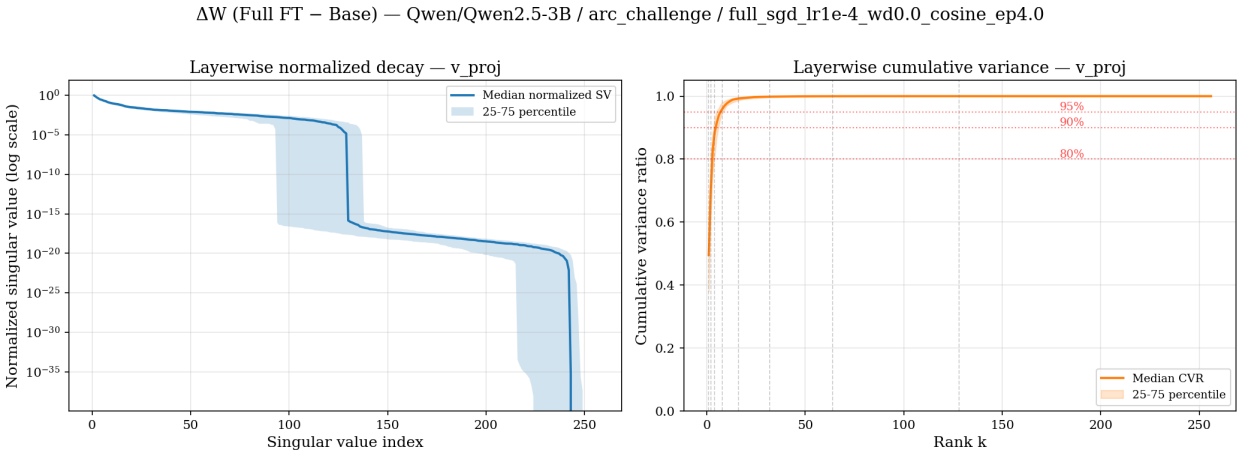

Figure 18: SVD of ΔW (Full FT − Base) — ARC-Challenge (lr=1e-4), V projection.